Team 4- Final Presentation

RIT LEARNER PARTICIPATION ANALYSIS

Opportunity Demand: Findings & Recommendations

Historical performance • seasonal & category demand patterns • forecasted demand • model performance • actionable recommendations

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Prepared for Program Stakeholders | Data: Jan 2023 – Apr 2024 | 8,558 applications across 23 opportunities

EXECUTIVE SUMMARY

The story in five numbers

8,558

Total applications

Jan 2023 – Apr 2024

47.5%

Overall participation rate

21.5%

Internship conversion

Least conversion rate vs Signup volume(63%)

2,755

Peak quarter volume

Q1 2024 sign-ups

2,450–2,520

Forecasted quarterly volume

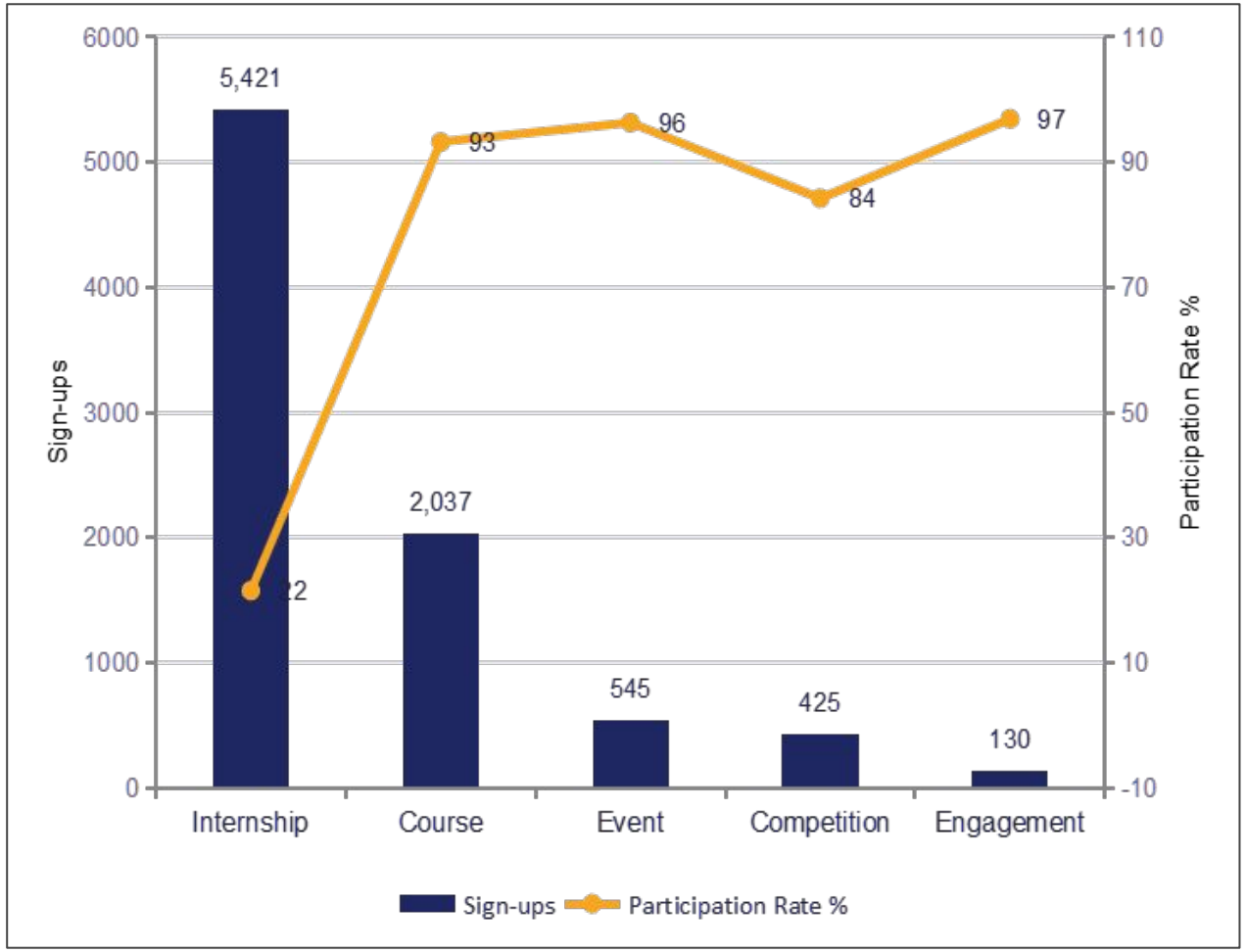
5 Categories; 3 Models

Opportunity Category	Total Applications	Participated	Participation Rate
Internship	5421	1165	**21.5%**
Course	2037	1898	**93.2%**
Event	545	525	**96.3%**
Competition	425	358	**84.2%**
Engagement	130	126	**96.9%**

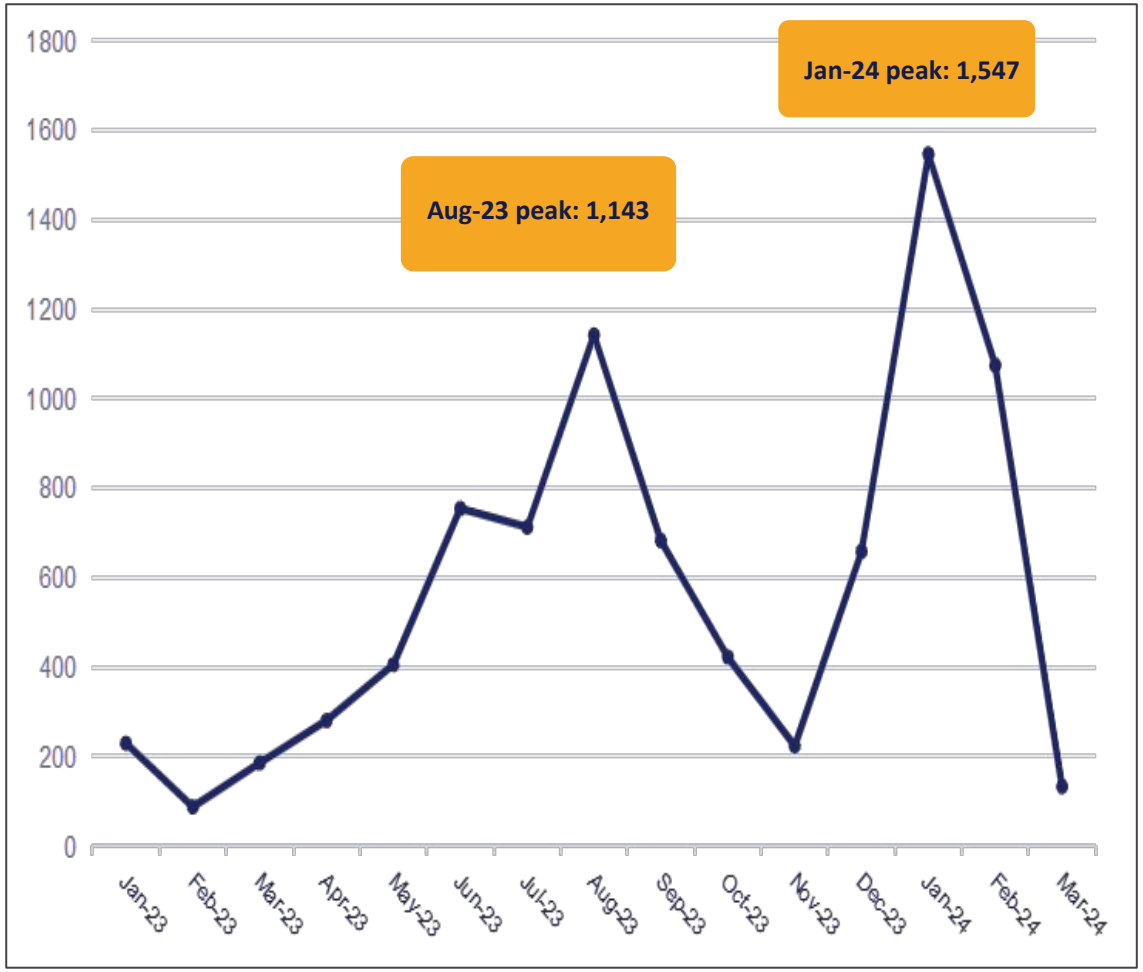


Internships dominate volume; signups are sharp seasonally

Sign-ups & participation rate by category



Monthly sign-up volume



Both peaks align with the start of academic terms in the US and India — the platform's two largest source countries (46% and 33% of applicants respectively).

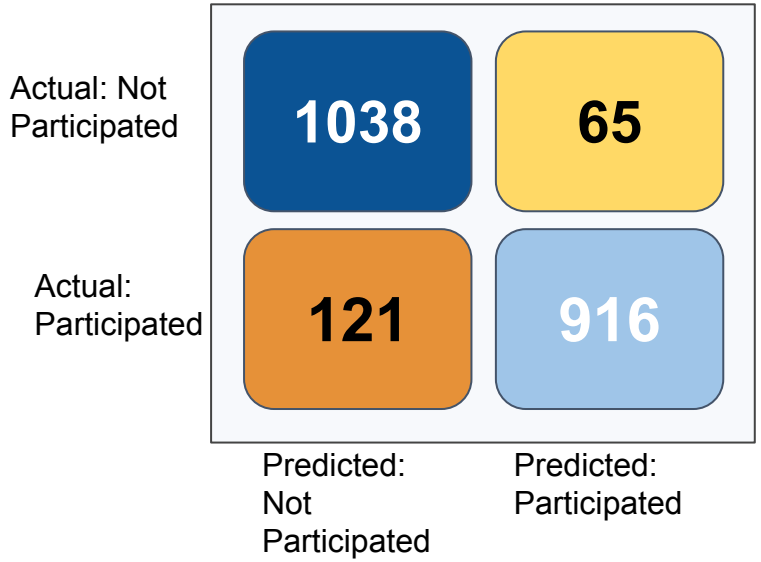
The prediction model separates participants from non-participants with 91% accuracy

Model prediction performance comparison: Gradient Boosting, Random Forest & Logistic Regression

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Gradient Boosting	91.3%	93.4%	88.3%	90.8%	0.970
Random Forest	91.1%	92.0%	89.3%	90.7%	0.968
Logistic Regression	81.95	82.7%	79.5%	81.0%	0.920

Gradient Boosting is the best performer and is used for the projections in this deck. Precision is very high (93%) - when the model predicts "will participate," it is rarely wrong - while recall is more moderate (88%), reflecting the genuinely competitive, not-fully-predictable nature of Internship selection. Implying internship participation is dependent on other unpredictable factors not a model failure.

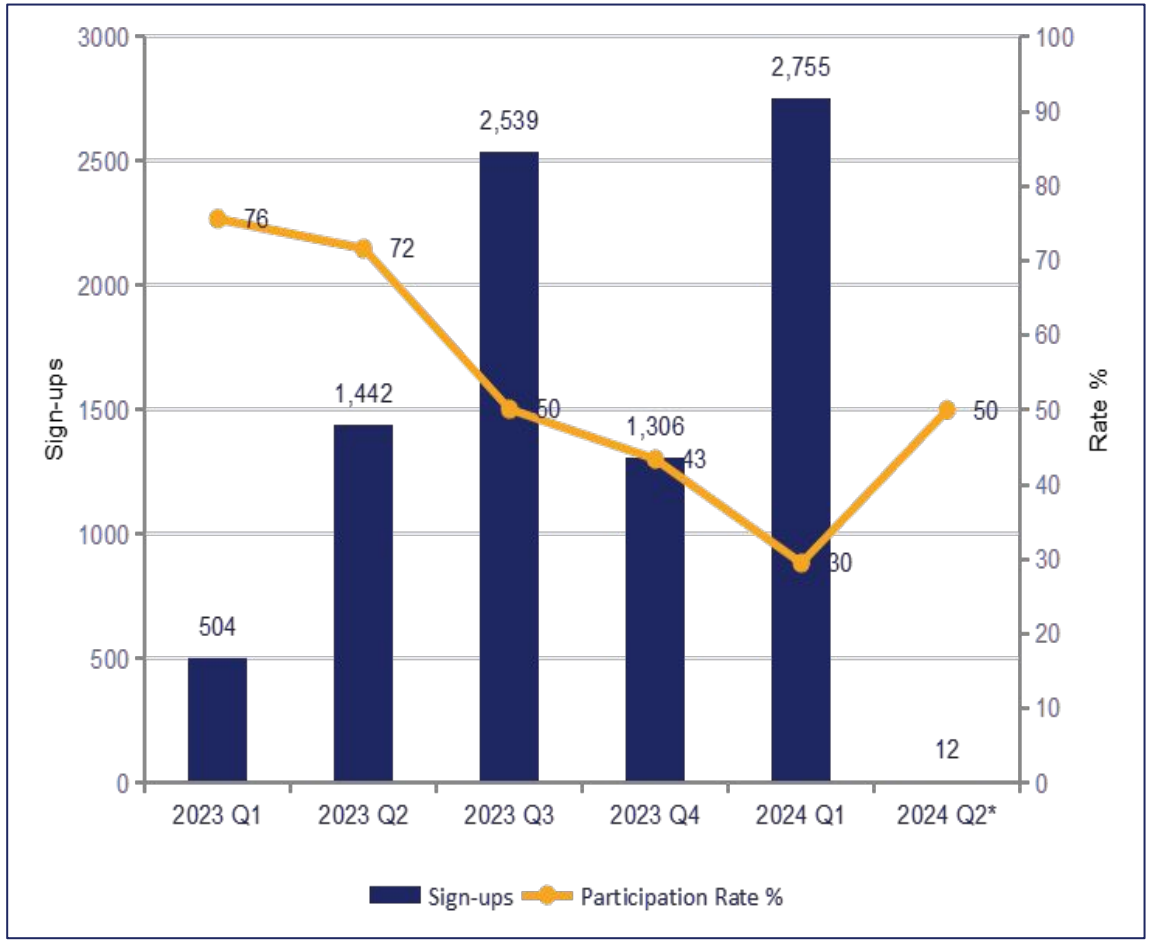
Confusion Matrix- Gradient Boosting



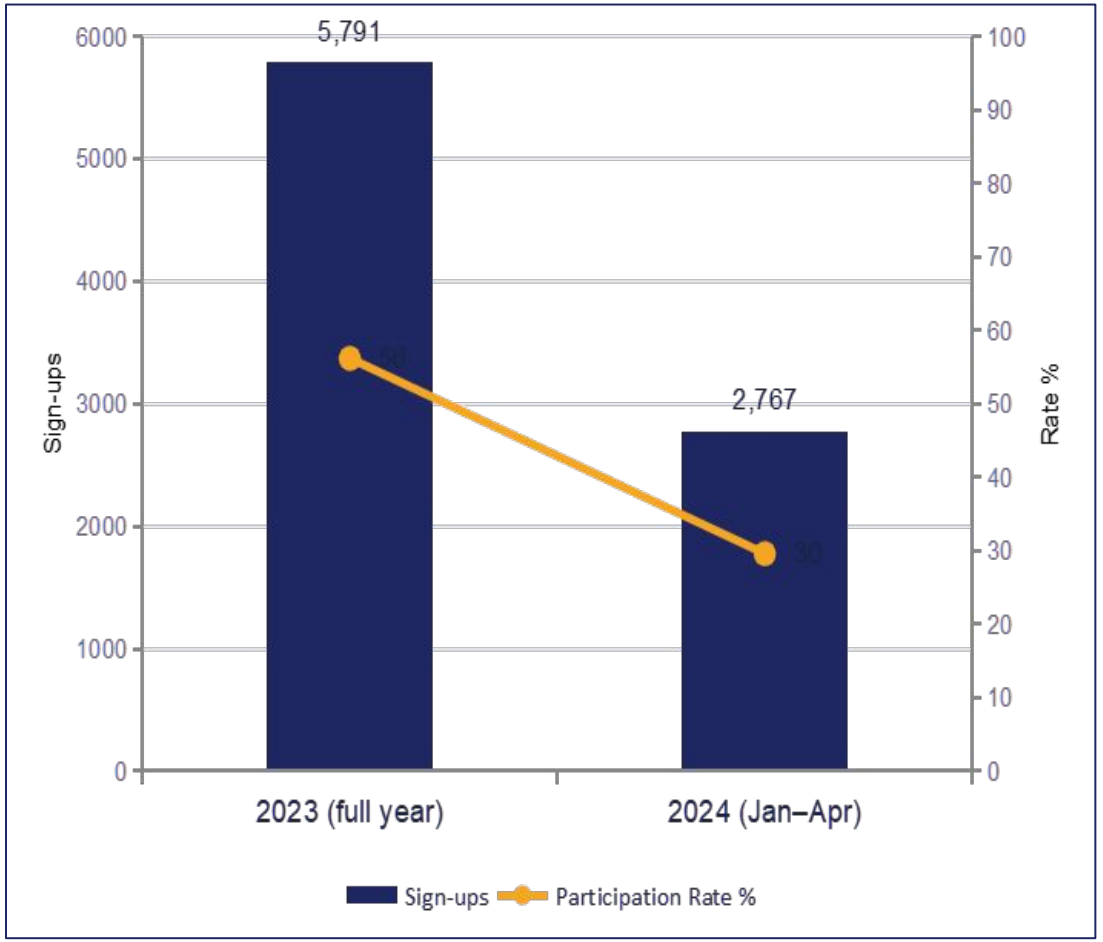
The best model classified 1954(91%) correctly. Most errors(121) are False negatives - predicted 'did not participate' for a student who actually participated.

Quarterly volume is climbing;Yearly volume is on the rise(2024(Jan-Apr)=48% of 2023)

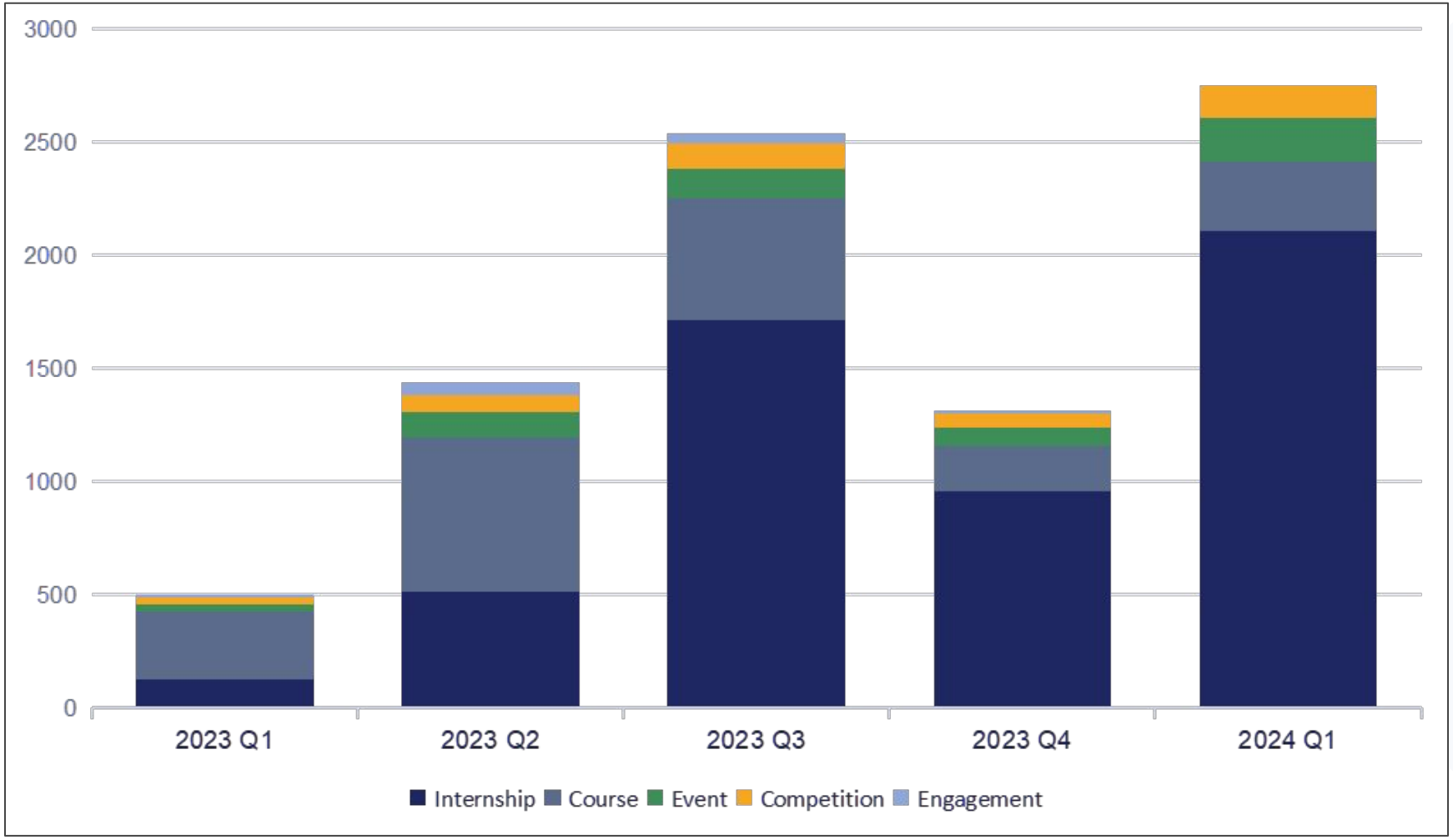
Quarter sign-ups volume vs participation rate



Year-to-year: volume vs participation rate



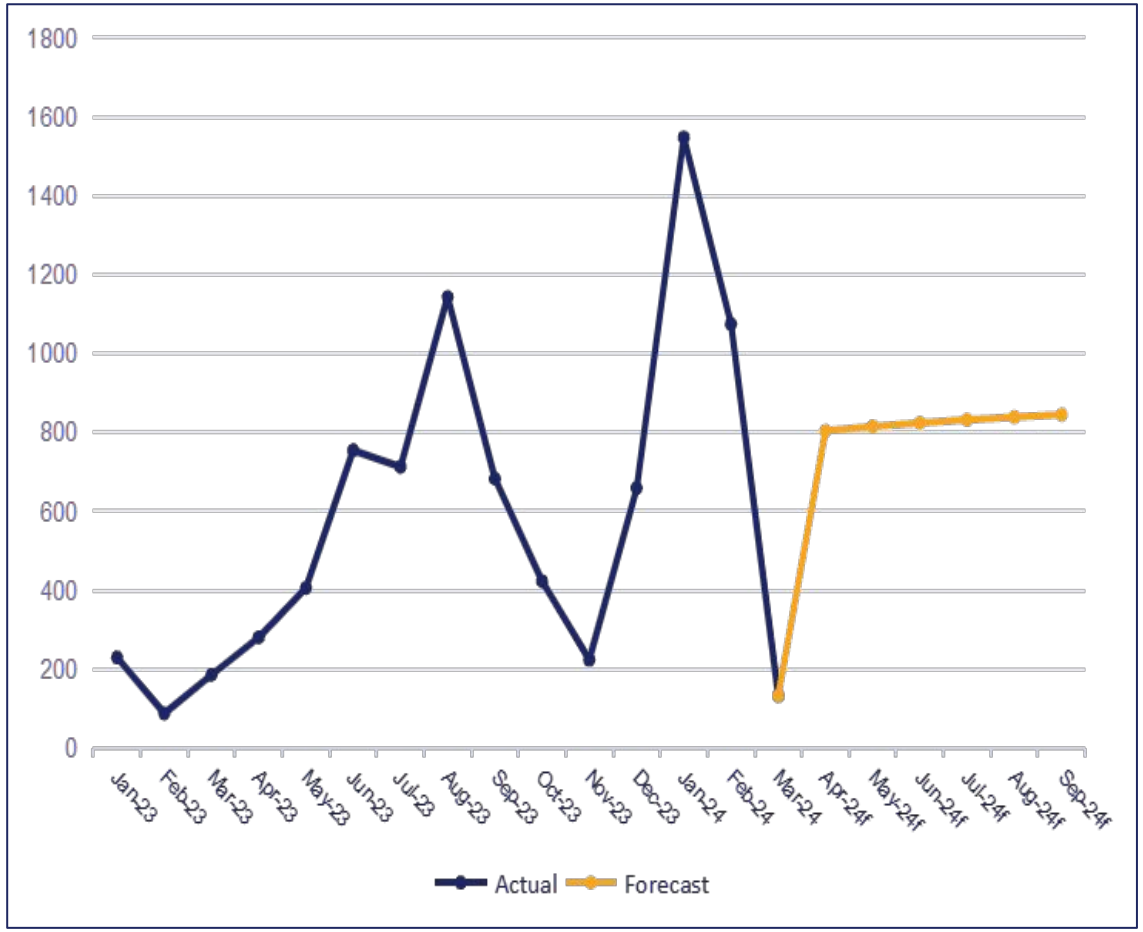
The demand mix is shifting toward Internships each quarter



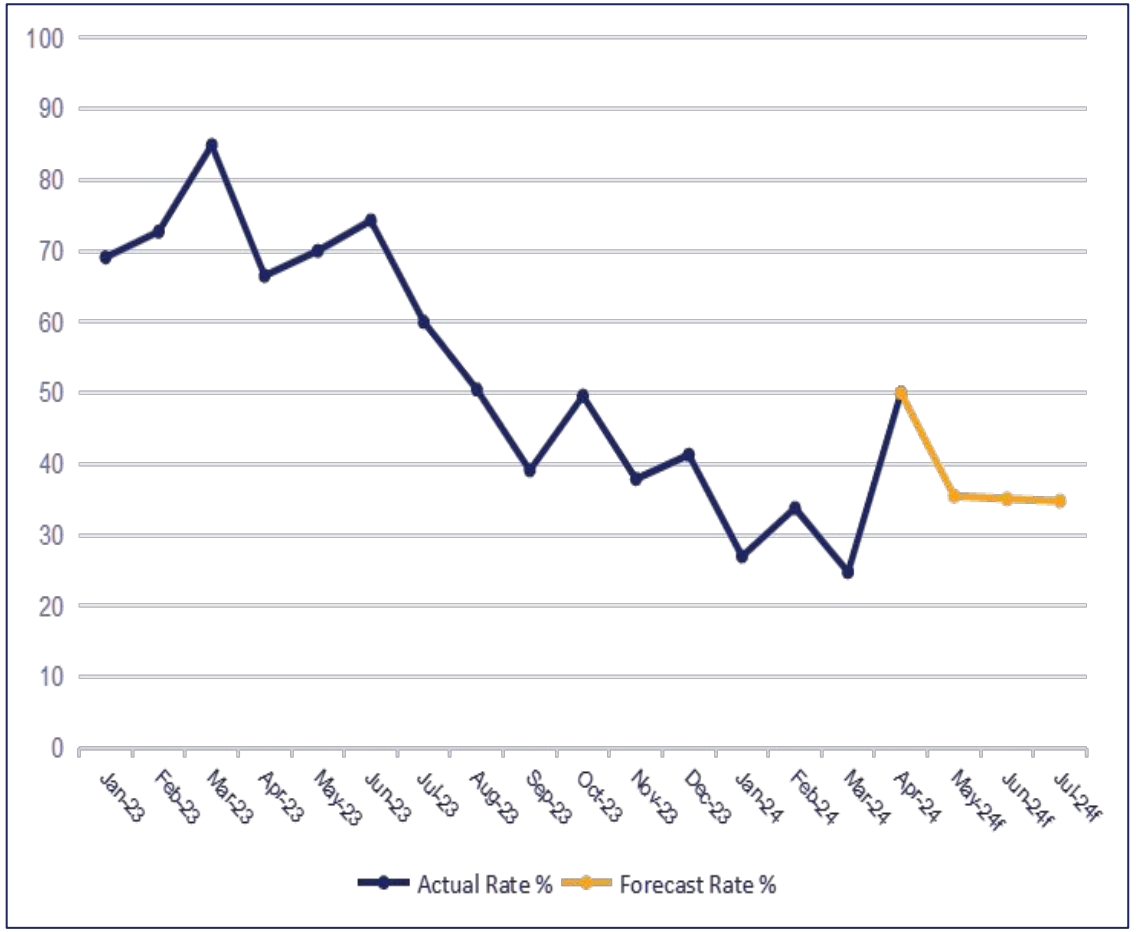
- Internship's share of quarterly sign-ups grew from 25% (Q1 2023) to 76% (Q1 2024).
- Engagement fell to zero in Q1 2024 — no new opportunities appear to have been posted since Nov 2023.
- Course volume, the second-largest category, has been roughly flat to declining since its Q2 2023 high.

Volume is projected to normalize toward 800–845 sign-ups/month

Sign Up Volume: Actual vs Forecasted



Participation rate: Actual vs Forecasted

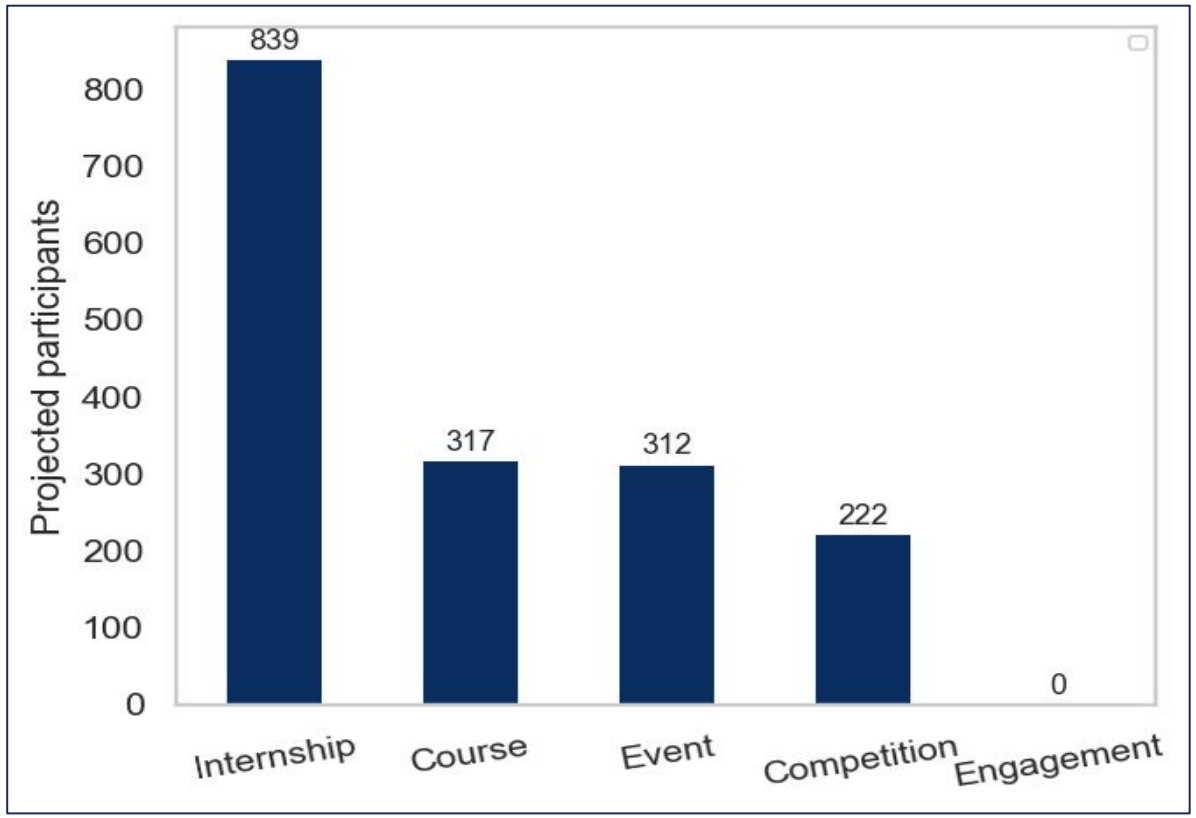


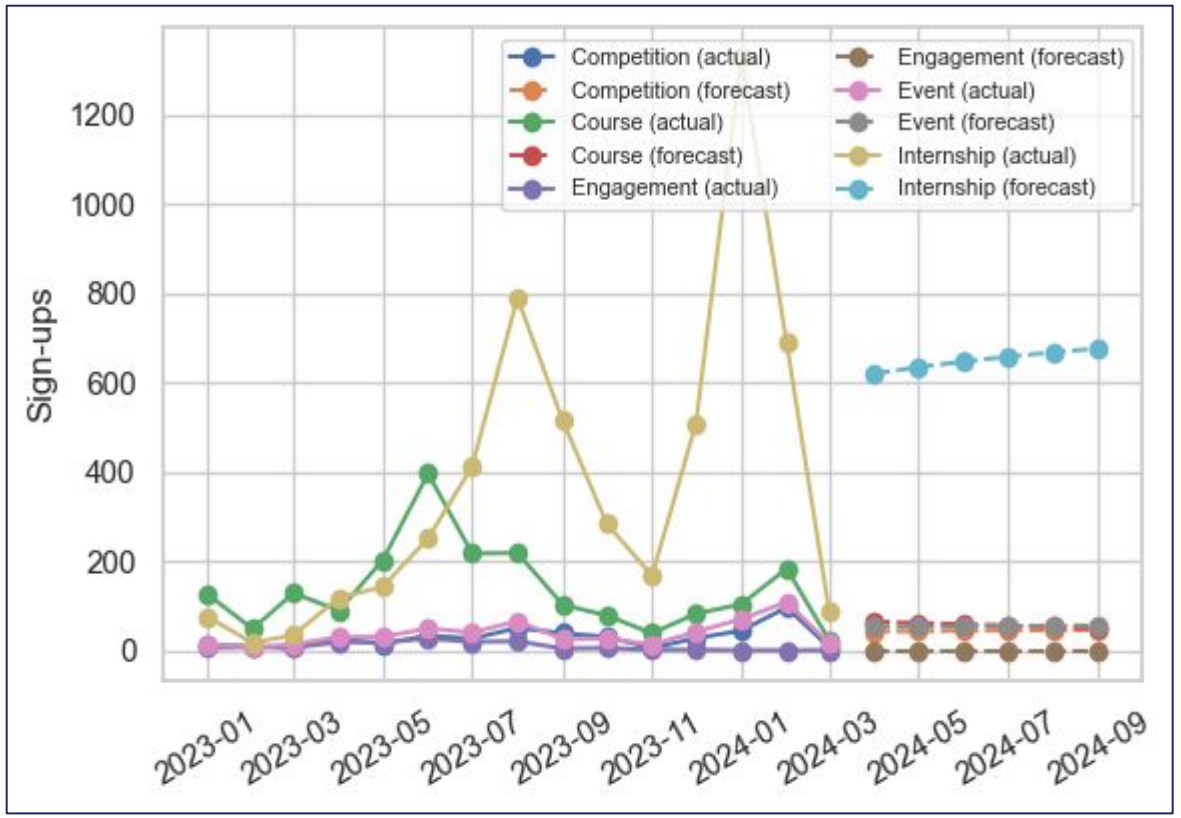
Damped Holt linear-trend model fit on 15 complete months (Apr 2024 excluded as a partial month). Wide month-to-month volatility means this is a directional planning estimate — refresh monthly as new data lands.

3. FORECASTED DEMAND

Internships will keep producing the most participants despite the lowest conversion rate

Projected next-6-months sign-ups vs. actual participants, by category

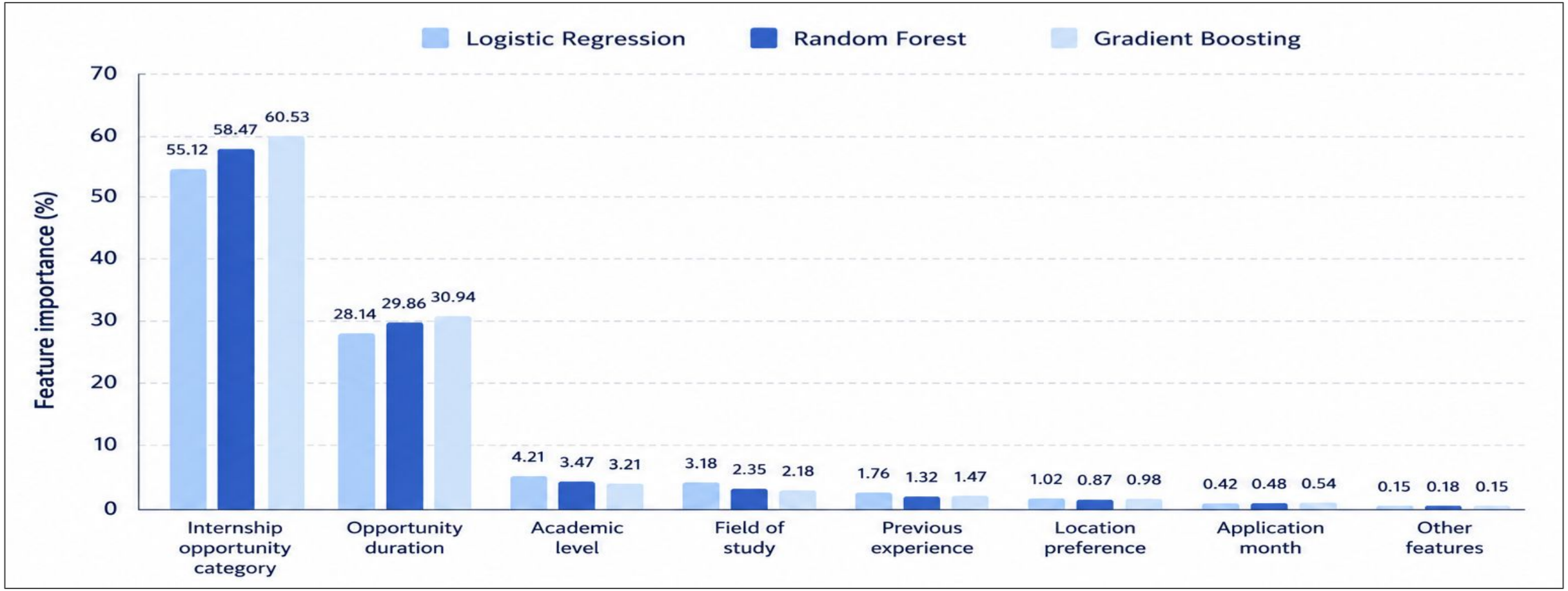




Even though Internships convert at the lowest rate, their sheer volume means they will continue to generate the largest absolute number of actively participating learners of any category — roughly 2.5x the next-largest category (Course).

One Feature dominates- whether the opportunity is an ‘internship’ for each model

Higher values indicate greater weight of the features in predicting participation



The models predictive power comes from whether the opportunity is an internship or not(**60%- GB model**). Opportunity duration days is a secondary signal, while everything else has medium to high negligible impact

ACTIONABLE RECOMMENDATIONS

Where to invest resources, outreach target, and when to intake

Resource Allocation — where to invest

1. Staff Internship review to its screening load
63.3% of all applications resolve through active rejection (65.84%) rather than simple allocation — review capacity should follow this volume, not the conversion rate.

2. Fund a deep-dive into Health Care Management
The only Internship converting above 50% (54.85% vs. 11–22% elsewhere) — a focused study of what it does differently is low-cost and high-return.

3. Split model output by function, not one blanket use
Random Forests list (fewer missed participants, 110 vs. 121) to outreach; Gradient Boosting list (fewer over-promised slots, 65 vs. 80) to capacity planners.

4. Redirect demographic-targeting budget
Country/gender carry near-zero predictive weight (0.0057 / 0.0013) — move that spend into redesigning the Internship process itself.

Promotion Gaps — where the opportunity is

1. Non-STEM applicants are under-promoted for Internships
Convert 1.5x higher than STEM (29.7% vs. 19.6%) but are only 19% of Internship applicants (1,026 of 5,417) — the clearest promotion gap on the platform.

2. Engagement is nearly invisible despite converting best
96.9% positive — the highest of any category — but just 1.5% of all volume (130 of 8,558 records). The smallest, most under-promoted category on the platform.

3. Competition's Pending pool is a ready re-engagement list
14.6% of Competition applicants sit stuck at "Applied" — the largest pending share of any category — already-interested learners no one has followed up with.

4. Course isn't marketed as the Internship on-ramp
93.2% positive progression at real volume (2,037 records) is currently promoted as a separate option, not as the readiness step that leads into Internships.

ACTIONABLE RECOMMENDATIONS

Business Impact Assessment, Planning Benefits, Model Operationalization & Limitation

USE PARTICIPATION RATE AS PRIMARY BUSINESS ASSESSMENT MEASURE

- Participation rate tracked at opportunity category, major, country, and time period level rather than a single blended number.

PLANNING BENEFITS

- **CAPACITY PLANNING-** rely on the forecast for staffing review & resource allocation in advance.
- **BUDGET ALLOCATION -** splitting categories into “reach” vs “efficiency” roles as a defensible basis in finance proposals
- **MODEL RETRAIN FLEXIBILITY -** the model can be trained quarterly as new cohorts resolve

MODEL BENEFITS

- **EARLY WARNING TOOL -** the model predicts participation likelihood at the onset of the learner sign up.
- **REPEATABLE METHODOLOGY -** The Cleaning Pipeline, Feature Engineering, and Model can be re-run each quarter or any future period.

POTENTIAL ORGANIZATIONAL LIMITATION

- **SHORT HISTORY -** Limits forecasting , estimates need not be relied on precise targets
- **DATA QUALITY DEPENDENCY-** Some source fields(timestamps, Institution Names) require manual correction else it carries a residual risk of misclassification.
- **MODEL MAINTENANCE -** Without a retraining commitment,prediction quality will degrade.

Resources Links

- Presentation Video - (https://drive.google.com/file/d/1eUSWvxfEiDNLylXM4rG7vzlE1IXcprAx/view?usp=drive_link)
- AI Documentation doc - (AI DOCUMENTATION)
- Analysis Dashboard file - (RIT_ANALYSIS DASHBOARD)
- Machine learning notebook ipynb - (Notebook.pdf)
- Cleaned_dataset - (RIT_Student_Engag_Clean_CSV)

THE END!

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